LAB- 5

1. A → General Purpose Register
   B → Index Register
   C → Pointer Register
   D → Segment Register
   E → Flags

2. 1. Instruction Pointer
   2 Register
   3 Code Segment, Data Segment, Stack Segment
   4. Directives
   5 Big- endian
   6. AX, BX, CX, DX.

3. num C dw 3A5Bh.

num D dw $\underbrace{1001}_{9}\ \underbrace{1001}_{9}\ \underbrace{1001}_{9}\ \underbrace{0110}_{6}$

AX	AH	AL	
	3A	5B	mov ax, numC
BX	BH	BL	
	99	96	mov bx, numD

or ah, bl.

AH = 3Ah = 0011 1010

BL = 96h = 1001 0110

AH ← 0011 1010 OR 1001 0110

= $\underbrace{1011}_{B}\ \underbrace{1110}_{E}$

AH = BE

AX	AH	AL
	BE	5B
BX	BH	BL
	99	96

xchg ax, bx.

AX	AH	AL
	99	96
BX	BH	BL
	BE	5B

→ mov al, 30h.

AX	AH	AL
	99	30
BX	BH	BL
	BE	5B

→ mul bl

AX ← Al x Bl

```
5B →          0101  1011
30          x 0011  0000
            ---------------
        0101  1011  0000
      0101 1  0110  0000
      ---------------------
      0001  0001  0001  0000
      1     1     1     0
```

	AH	AL
AX	11	10
	BH	BL
BX	BE	5B

→ dec ax

```
AX =   0001  0001  000X  10101010
                         ~~~~~~~~
                                -1
      ---------------------------------
      0001,  0001,  0000,  1111
      1      1      0      F
```

	h	l
AX	11	0F
BX	BE	5B

→ mov cx , 7F 00h.

	h	l
AX	11	0F
BX	BE	5B
CX	7F	00.

→ Sub ax, cx

AX ← AX − CX

$$
\begin{array}{lcccc}
 & 0 & 11110 & 11110 & \\
AX & \cancel{1} & \cancel{0000} & \cancel{0001} & 0000 \quad 1111 \\
CX & - & 0111 & 1111 & 0000 \quad 0000 \\
\hline
 & & \underbrace{1001}_{9} & \underbrace{0010}_{2} & \underbrace{0000}_{0} \quad \underbrace{1111}_{F}
\end{array}
$$

	h	l
AX	92	0F
BX	BE	5B
CX	7F	00

$AX = 920F$

$= 9 \times 16^3 + 2 \times 16^2 + 0 \times 16^1 + 15 \times 16^0$

$= 36864 + 512 + 0 + 15$

$= 37391$

$BX = BE$

$= 11 \times 16^1 + 14 \times 16^0$

$= 176 + 14$

$= 190$

AX | BH

```
        196
190 | 37391
    - 190↓
      1839
      1710
       1291
       1140
        151
```

Remainder = 151

Quotient = 196

```
      12
16 | 196
   - 16↓
     36
     32
      4
```

196 = 12 in hex = C 4 remainder = C4

```
      9
16 | 151
   - 144
      7
```

151 = 97 in hex

	h	l
AX	97	C4
BX	BE	5B
CX	7F	00